PetStuff

--- A Full Stack MERN-Based Multi-Vendor E-Commerce Platform for Pet Owners ---

Project Type: Full Stack Web Application
Purpose: Learning MERN Stack & Developing My Portfolio
Technology Stack: MongoDB | Express.js | React.js | Node.js
Developer: Mohd Sayam

GitHub Repository: *Repo Link*
Video Demonstration: *(7–8 min walkthrough – Add link)*

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1. Executive Summary

PetStuff is a comprehensive **full-stack e-commerce platform** built specifically for pet owners.
The platform bridges the gap between **pet product vendors (store owners)** and **customers**, enabling a smooth and secure shopping experience for pet food, medicines, toys, and accessories.

Developed using the **MERN Stack (MongoDB, Express.js, React.js, Node.js)**, PetStuff emphasizes:

- High performance
- Secure authentication
- Modular architecture
- A modern, responsive user interface

The project demonstrates **real-world full-stack development practices**, including RESTful APIs, role-based access control, cloud media management, and scalable system design.

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2. Project Vision & Novelty

Unlike generic e-commerce platforms, PetStuff is designed with a **Multi-Vendor Architecture** at its core.

The platform clearly distinguishes between:

- **Customers** (pet owners)
- **Store Owners (Admins)**

This separation allows multiple vendors to operate independently within a single ecosystem, making the application **scalable and marketplace-ready**.

Key Unique Selling Propositions (USPs)

- **Pet-Centric Categorization**
  Products are intelligently categorized by animal type (Dog, Cat, Bird, etc.), enabling faster and more relevant discovery.
- **Dynamic Pricing Engine**
  Sale prices and discount percentages are automatically calculated, ensuring consistency and reducing manual errors.

- **Hybrid Authentication System**
  Users can authenticate via:
    - Email & Password
    - Google OAuth (one-click login)
- **Vendor Autonomy**
  Each store owner gets a **dedicated dashboard** to manage products, orders, and analytics without affecting other vendors or global platform settings.

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3. Technology Stack

3.1 Frontend (Client-Side)

- **Framework:** React.js (Vite)
  Chosen for fast development builds and optimized performance.
- **Styling:** Tailwind CSS
  Enables a mobile-first, responsive, and consistent UI design.
- **State Management:** React Context API
  Used for managing global states such as:
    - Authentication
    - Shopping Cart
    - UI Theme
- **Network Layer:** Axios
  Configured with interceptors for secure token-based API communication.
- **UX Enhancements:**
    - `react-hot-toast` for real-time user feedback
    - `lucide-react` for clean and modern iconography

3.2 Backend (Server-Side)

- **Runtime & Framework:** Node.js with Express.js
  Provides scalable and maintainable RESTful API services.
- **Database:** MongoDB (NoSQL)
  Enables flexible schema design and high-performance queries.
- **Authentication & Security:**
    - JSON Web Tokens (JWT) for session handling
    - Passport.js for Google OAuth authentication strategies
- **Media Management:**
    - Cloudinary for optimized image storage and delivery

- **Email Communication:**
    - Nodemailer for transactional emails such as:
        - Email verification
        - Password reset flows

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4. Detailed Features

4.1 Customer Features

- **Seamless Onboarding**
  Simple sign-up process with support for Google OAuth.
- **Smart Search & Product Discovery**
  Filter products by:
    - Animal type (Dog, Cat, etc.)
    - Product category (Food, Toys, Medicines)
- **Shopping Cart System**
  Persistent cart allowing:
    - Quantity updates
    - Real-time price calculation
- **Secure Checkout**
  Order placement with address management and order summary.
- **Order Tracking**
  Real-time order status updates:
  Processing → Shipped → Delivered
- **Profile Management**
  Secure handling of personal details and password recovery.

4.2 Store Owner (Admin) Features

- **Store Creation**
  Automated setup of a digital storefront with location and descriptive metadata.
- **Product Lifecycle Management**
  Full **CRUD operations**:
    - Add new products
    - Update pricing and stock
    - Delete discontinued items
- **Order Fulfillment System**
  Store owners can view and manage only the orders related to their products.

- **Analytics Dashboard**
  Sales and order insights visualized using **Recharts**, enabling data-driven decisions.

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5. Database Architecture & Entity Relationships

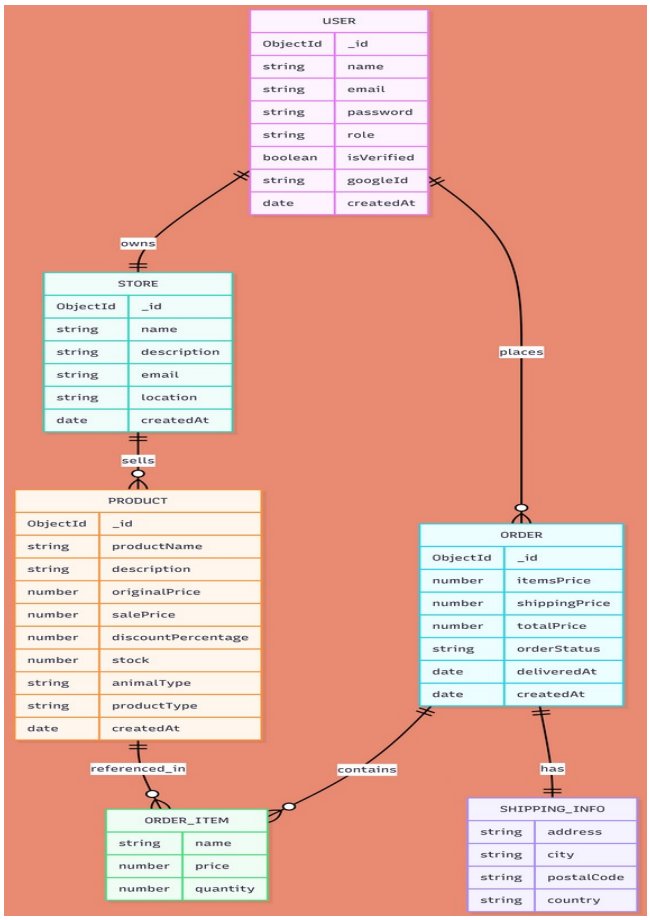
Description:

The PetStuff platform uses a structured MongoDB schema design to maintain relational integrity across users, stores, products, and orders.

The database follows a normalized approach where:

- Users can act as Customers or Store Owners
- Each Store is owned by a single Admin
- Products belong to specific Stores
- Orders maintain immutable product snapshots to ensure historical accuracy

The following ER diagram represents the complete database structure and relationships used in the system.



6. Functional Workflows & Highlights

- **Authentication Flow**
  Email verification via tokenized links or instant login via Google OAuth.
- **Purchase Flow**
  Product added → Checkout → Stock deduction → Order creation → Store owner notification.
- **Deployment Strategy**
  Implements a **“Chicken & Egg” deployment pattern on Vercel** to resolve circular environment variable dependencies between frontend and backend securely.

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7. Future Roadmap

- Integration of **Stripe / Razorpay** payment gateways
- Product **review and rating system**
- **Real-time chat** between store owners and customers
- Advanced analytics and performance optimizations
- Mobile application version

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8. Video Demonstration

A **7–8 minute detailed walkthrough video** is attached with this documentation, explaining:

- Project motivation
- UI walkthrough
- Authentication flow
- Product & order management
- Backend architecture and APIs

The video is designed to be **concise, engaging, and well explained.**

Video Link: Video